P2 Chapter 9: Differentiation

Trigonometric Gradient Functions

Differentiating other trigonometric functions

Differentiate $y = \tan x$

$$y = \frac{\sin x}{\cos x}$$

Use definition of $\tan x$

$$u = \sin x \qquad v = \cos x$$

Use quotient rule.

$$\frac{du}{dx} = \cos x \qquad \frac{dv}{dx} = -\sin x$$

$$\frac{dy}{dx} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

✐ More generally: $\frac{d}{dx}(\tan kx) = k \sec^2 kx$

Tip: This is in the formula booklet, but I HIGHLY encourage your to memorise it to save you time in the exam.

Differentiate $y = \sec x$

$$y = \frac{1}{\cos x} = (\cos x)^{-1}$$

Tip: We could have always used the quotient rule on $\frac{1}{\cos x}$. However, I tend to the find the chain rule is marginally simpler if the numerator is a constant.

$$\frac{dy}{dx} = -(\cos x)^{-2} \times (-\sin x)$$

$$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \times \frac{\sin x}{\cos x} = \sec x \tan x$$

✐ More generally: $\frac{d}{dx}(\sec kx) = k \sec kx \tan kx$

Differentiating other trigonometric functions

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\cot x) = -\mathrm{cosec}^2\, x$$

$$\frac{d}{dx}(\mathrm{cosec}\, x) = -\,cosec\, x \cot x$$

Memory Tip: To memorise the two on the right, notice that when we 'co' the two on the left, it 'co's each term in the derivative, but also negates it.

Differentiate

(a) $y = \frac{cosec\ 2x}{x^2}$

(b) $y = \sec^3 x$

a

?

b

?

Differentiating other trigonometric functions

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\cot x) = -\text{cosec}^2\, x$$

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Memory Tip: To memorise the two on the right, notice that when we 'co' the two on the left, it 'co's each term in the derivative, but also negates it.

Differentiate

(a) $y = \frac{cosec\ 2x}{x^2}$

(b) $y = \sec^3 x$

a

$u = cosec\ 2x \quad v = x^2$

$\frac{du}{dx} = -2cosec\ 2x \cot 2x \quad \frac{dv}{dx} = 2x$

$$\frac{dy}{dx} = \frac{-2x^2 cosec\ 2x \cot 2x - 2x\ cosec\ 2x}{x^4}$$

b

$y = (\sec x)^3$

$\frac{dy}{dx} = 3(\sec x)^2 \times \sec x \tan x$

$= 3 \sec^3 x \tan x$

Observation: Notice that when differentiating a power of sec, the power stays the same. This will be relevant when we come to integration.

Test Your Understanding So Far

Edexcel C3 June 2013(R) Q5b

(*b*) Show that $\frac{d}{dx}(\sec^2 3x)$ can be written in the form

$$\mu(\tan 3x + \tan^3 3x)$$

where μ is a constant. **(3)**

?

Test Your Understanding So Far

Edexcel C3 June 2013(R) Q5b

(*b*) Show that $\frac{d}{dx}(\sec^2 3x)$ can be written in the form

$$\mu(\tan 3x + \tan^3 3x)$$

where μ is a constant. **(3)**

$$\frac{d}{dx}\left((\sec 3x)^2\right) = 2\sec 3x \times 3\sec 3x \tan 3x$$
$$= 6\sec^2 3x \tan 3x$$
$$= 6(1 + \tan^2 3x)\tan 3x$$
$$= 6(\tan 3x + \tan^3 3x)$$

Making use of $1 \div dx/dy$

Often in exam questions, you will be given x in terms of y, but want to find $\frac{dy}{dx}$ in terms of x.

The key is to make use of an appropriate trig identity, e.g:

$$\boldsymbol{\sin^2 x + \cos^2 x \equiv 1 \qquad 1 + \tan^2 x \equiv \sec^2 x}$$

Given that $x = \tan y$, express $\frac{dy}{dx}$ in terms of x.

$$\frac{dx}{dy} = \sec^2 y$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

Note that $1 + \tan^2 y \equiv \sec^2 y$

$$= \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

You might be tempted to put $\frac{dy}{dx} = \cos^2 y$, but there's not a relationship between $\cos^2 y$ and $\tan y$ that would enable us to later get x into the equation.

Using $1 + \tan^2 y \equiv \sec^2 y$ is sensible because we know that $x = \tan y$.

Further Examples

Show that if $y = \arcsin x$, then $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

?

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

Given that $y = \arcsin x^2$ find $\frac{dy}{dx}$

?

Further Examples

Show that if $y = \arcsin x$, then $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

$$\boldsymbol{y = \arcsin x}$$
$$\boldsymbol{x = \sin y}$$
$$\boldsymbol{\frac{dx}{dy} = \cos y}$$
$$\boldsymbol{\frac{dy}{dx} = \frac{1}{\cos y}}$$

Remember that we're trying to turn $\cos y$ into an expression in terms of x; we have to use $x = \sin y$ in some way. You then might think "Oh, I know an identity that relates $\cos y$ and $\sin y$!"

We know that $\boldsymbol{sin^2 y + cos^2 y \equiv 1}$

$$\boldsymbol{= \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}}$$

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

Given that $y = \arcsin x^2$ find $\frac{dy}{dx}$

Using the chain rule:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x^2)^2}} \times 2x$$
$$= \frac{2x}{\sqrt{1-x^4}}$$

Test Your Understanding

Edexcel C3 Jan 2011 Q8b,c

Given that

$$x = \sec 2y,$$

(*b*) find $\frac{dx}{dy}$ in terms of y. **(2)**

(*c*) Hence find $\frac{dy}{dx}$ in terms of x. **(4)**

?

[Textbook] Given that $y = \arctan\left(\frac{1-x}{1+x}\right)$, find $\frac{dy}{dx}$

?

Test Your Understanding

Edexcel C3 Jan 2011 Q8b,c

Given that

$$x = \sec 2y,$$

(*b*) find $\frac{dx}{dy}$ in terms of y. **(2)**

(*c*) Hence find $\frac{dy}{dx}$ in terms of x. **(4)**

(b)	$x = \sec 2y, \quad y \neq (2n+1)\frac{\pi}{4}, n \in \mathbb{Z}.$		
	$\frac{dx}{dy} = 2\sec 2y \tan 2y$	$K\sec 2y \tan 2y$ $2\sec 2y \tan 2y$	M1 A1
			(2)
(c)	$\frac{dy}{dx} = \frac{1}{2\sec 2y \tan 2y}$	Applies $\frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)}$	M1
	$\frac{dy}{dx} = \frac{1}{2x \tan 2y}$	Substitutes x for $\sec 2y$.	M1
	$1 + \tan^2 A = \sec^2 A \Rightarrow \tan^2 2y = \sec^2 2y - 1$	Attempts to use the identity $1 + \tan^2 A = \sec^2 A$	M1
	So $\tan^2 2y = x^2 - 1$		
	$\frac{dy}{dx} = \frac{1}{2x\sqrt{(x^2-1)}}$	$\frac{dy}{dx} = \frac{1}{2x\sqrt{(x^2-1)}}$	A1
			(4)

[Textbook] Given that

$y = \arctan\left(\frac{1-x}{1+x}\right)$, find $\frac{dy}{dx}$

(Let's use the 'full' Chain Rule method this time)

Let $u = \frac{1-x}{1+x}$

$$\frac{du}{dx} = \cdots = -\frac{2}{(1+x)^2}$$

$$y = \arctan u$$

$$\therefore \frac{dy}{du} = \frac{1}{1+u^2}$$

By the Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{1+u^2} \times \left(-\frac{2}{(1+x)^2}\right)$$

$$= \cdots$$

$$= -\frac{1}{1+x^2}$$

Exercise 9.6

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Extension:

[STEP I 2011 Q3] Prove the identity

$$4\sin\theta\sin\left(\frac{1}{3}\pi-\theta\right)\sin\left(\frac{1}{3}\pi+\theta\right)=\sin 3\theta \qquad (*)$$

(i) By differentiating (*) or otherwise, show that

$$\cot\frac{1}{9}\pi-\cot\frac{2}{9}\pi+\cot\frac{4}{9}\pi=\sqrt{3}$$

(ii) By setting $\theta=\frac{1}{6}\pi-\phi$ in $(*)$, or otherwise, obtain a similar identity for $\cos 3\theta$ and deduce that

$$\cot\theta\cot\left(\frac{1}{3}\pi-\theta\right)\cot\left(\frac{1}{3}\pi+\theta\right)=\cot 3\theta$$

Show that

$$cosec\,\frac{1}{9}\pi-cosec\,\frac{5}{9}\pi+cosec\,\frac{7}{9}\pi=2\sqrt{3}$$

Homework Exercise

1 Differentiate:

a $y = \tan 3x$ **b** $y = 4\tan^3 x$ **c** $y = \tan(x-1)$ **d** $y = x^2 \tan \frac{1}{2}x + \tan\left(x - \frac{1}{2}\right)$

2 Differentiate:

a $\cot 4x$ **b** $\sec 5x$ **c** $\operatorname{cosec} 4x$ **d** $\sec^2 3x$

e $x \cot 3x$ **f** $\dfrac{\sec^2 x}{x}$ **g** $\operatorname{cosec}^3 2x$ **h** $\cot^2(2x-1)$

3 Find the function $f'(x)$ where $f(x)$ is:

a $(\sec x)^{\frac{1}{2}}$ **b** $\sqrt{\cot x}$ **c** $\operatorname{cosec}^2 x$ **d** $\tan^2 x$

e $\sec^3 x$ **f** $\cot^3 x$

4 Find $f'(x)$ where $f(x)$ is:

a $x^2 \sec 3x$ **b** $\dfrac{\tan 2x}{x}$ **c** $\dfrac{x^2}{\tan x}$ **d** $e^x \sec 3x$

e $\dfrac{\ln x}{\tan x}$ **f** $\dfrac{e^{\tan x}}{\cos x}$

5 The curve C has equation

$$y = \frac{1}{\cos x \sin x}, \quad 0 < x \leqslant \pi$$

a Find $\dfrac{dy}{dx}$ **(4 marks)**

b Determine the number of stationary points of the curve C. **(2 marks)**

c Find the equation of the tangent at the point where $x = \dfrac{\pi}{3}$, giving your answer in the form $ax + by + c = 0$, where a, b and c are exact constants to be determined. **(3 marks)**

Homework Exercise

6 Show that if $y = \sec x$ then $\frac{dy}{dx} = \sec x \tan x$. **(5 marks)**

7 Show that if $y = \cot x$ then $\frac{dy}{dx} = -\text{cosec}^2 x$. **(5 marks)**

8 Assuming standard results for $\sin x$ and $\cos x$, prove that:

a the derivative of $\arccos x$ is $-\frac{1}{\sqrt{1-x^2}}$

b the derivative of $\arctan x$ is $\frac{1}{1+x^2}$

9 Given that $x = \text{cosec } 5y$,

a find $\frac{dy}{dx}$ in terms of y. **(2 marks)**

b Hence find $\frac{dy}{dx}$ in terms of x. **(4 marks)**

Homework Answers

1 **a** $3\sec^2 3x$ **b** $12\tan^2 x\sec^2 x$ **c** $\sec^2(x-1)$
d $\frac{1}{2}x^2\sec^2\frac{1}{2}x + 2x\tan\frac{1}{2}x + \sec^2\left(x-\frac{1}{2}\right)$

2 **a** $-4\operatorname{cosec}^2 4x$ **b** $5\sec 5x\tan 5x$
c $-4\operatorname{cosec} 4x\cot 4x$ **d** $6\sec^2 3x\tan 3x$
e $\cot 3x - 3x\operatorname{cosec}^2 3x$ **f** $\dfrac{\sec^2 x(2x\tan x - 1)}{x^2}$
g $-6\operatorname{cosec}^3 2x\cot 2x$
h $-4\cot(2x-1)\operatorname{cosec}^2(2x-1)$

3 **a** $\frac{1}{2}(\sec x)^{\frac{1}{2}}\tan x$ **b** $-\frac{1}{2}(\cot x)^{-\frac{1}{2}}\operatorname{cosec}^2 x$
c $-2\operatorname{cosec}^2 x\cot x$ **d** $2\tan x\sec^2 x$
e $3\sec^3 x\tan x$ **f** $-3\cot^2 x\operatorname{cosec}^2 x$

4 **a** $2x\sec 3x + 3x^2\sec 3x\tan 3x$
b $\dfrac{2x\sec^2 2x - \tan 2x}{x^2}$ **c** $\dfrac{2x\tan x - x^2\sec^2 x}{\tan^2 x}$
d $e^x\sec 3x(1 + 3\tan 3x)$ **e** $\dfrac{\tan x - x\sec^2 x\ln x}{x\tan^2 x}$
f $e^{\tan x}\sec x(\tan x + \sec^2 x)$

5 **a** $\dfrac{1}{\cos^2 x} - \dfrac{1}{\sin^2 x}$ **b** 2
c $24x - 9y + 12\sqrt{3} - 8\pi = 0$

6 $y = \dfrac{1}{\cos x}$, $\dfrac{dy}{dx} = \dfrac{\cos x \times 0 - 1 \times -\sin x}{\cos^2 x} = \dfrac{\sin x}{\cos^2 x}$
$= \sec x\tan x$

7 $y = \dfrac{1}{\tan x}$

$$\frac{dy}{dx} = \frac{\tan x \times 0 - 1 \times \sec^2 x}{\tan^2 x} = -\frac{\sec^2 x}{\tan^2 x} = -\frac{\frac{1}{\cos^2 x}}{\frac{\sin^2 x}{\cos^2 x}} = -\operatorname{cosec}^2 x$$

8 **a** Let $y = \arccos x \Rightarrow \cos y = x \Rightarrow \dfrac{dx}{dy} = -\sin y$

$$\frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-\cos^2 y}} = -\frac{1}{\sqrt{1-x^2}}$$

b Let $y = \arctan x$
Then, $\tan y = x$

$$\frac{dx}{dy} = \sec^2 y$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$$

9 **a** $\dfrac{-1}{5\cot 5y\operatorname{cosec} 5y}$ **b** $-\dfrac{1}{5x\sqrt{x^2-1}}$